System Components and Design

Purpose

• DriverPass is the client and wants a web-based system that supports driver training, online classes, practice tests, customer accounts, and on-the-road driving lesson reservations.
• The system should allow DriverPass staff and customers to manage relevant information while giving management access to reports and activity information.

System Background

• DriverPass was created to provide better preparation for students taking driving tests at the DMV through online classes, practice tests, and on-the-road training.
• The system needs customer accounts, package management, lesson reservations, driver and vehicle assignments, online testing, driver notes, DMV updates, reporting, and role-based access.
• DriverPass wants information available online from computers and mobile devices. Data changes must be made while online to avoid conflicting copies of data.

Objectives and Goals

• Provide customers with access to online course material and practice tests when included with their package.
• Allow customers and the secretary to create, cancel, and modify two-hour driving lesson reservations.
• Track the customer, package, lesson time, assigned driver, and assigned vehicle for each driving appointment.
• Provide activity tracking so DriverPass can identify who created, modified, or canceled a reservation and produce activity reports.
• Keep training tests and practice material current when the DMV provides new rules, policies, or sample questions.
• Provide role-based access and administrative account controls for DriverPass employees.

Requirements

Nonfunctional Requirements

Performance Requirements

• The system should be web-based and preferably hosted in the cloud, allowing authorized users to access the system from computers or mobile devices.
• The system should provide normal account, scheduling, and reporting functions without unnecessary delays during routine use.
• The system should support timely updates to DMV-related rules, policies, and sample questions and notify DriverPass when updates are received.
• Data should be centrally maintained online; users should not be able to modify offline copies in a way that could create duplicate or conflicting records.

Platform Constraints

• The system should operate through a web interface and preferably use cloud hosting.
• The back end should use a centralized database capable of storing customer, account, package, reservation, driver, vehicle, testing, and activity information.
• The system should support access from standard computers and mobile devices through a web browser.

Accuracy and Precision

• Each account should have identifiable user information and an assigned role so permissions can be applied correctly.
• Reservation records should accurately associate a customer with a two-hour lesson, date and time, driver, and vehicle.
• Customer registration should capture the required personal, contact, payment, pickup, and drop-off information.
• Activity records should identify who made, changed, or canceled a reservation so reports can be traced to the responsible user.

Adaptability

• The system should allow the IT officer to add, remove, modify, reset, or block user accounts without changing application source code.
• The system should allow an existing package to be disabled so new customers cannot register for that package.
• Adding or removing entire training modules is identified as a future enhancement and may require a developer or systems analyst.
• The cloud/web architecture should make it possible to maintain the system as supported platforms and technologies change.

Security

• The system should require authenticated accounts and apply different rights based on the user's role.
• The IT officer should have sufficient administrative access to reset passwords and block access for former employees.
• Data exchanged between users and the server should be protected through secure web communication and appropriate access controls.
• The system should record account and reservation activity so unauthorized or unexpected changes can be investigated.
• A secure account-recovery process should allow users to automatically reset a forgotten password.
• A brute-force protection policy should be included; the interview does not specify the exact lockout threshold, so that value should be finalized during detailed design.

Functional Requirements

• The system shall allow customers to create and maintain accounts.
• The system shall store customer first name, last name, address, phone number, state, credit card number, expiration date, security code, pickup location, and drop-off location.
• The system shall allow customers to schedule, cancel, and modify driving lesson appointments online.
• The system shall allow the secretary to create and manage appointments for customers.
• The system shall schedule driving sessions in two-hour blocks.
• The system shall associate each driving appointment with the appropriate customer, driver, vehicle, date, and time.

• The system shall support the three DriverPass training packages and allow an administrator to disable a package.
• The system shall provide online course content and practice tests for customers whose package includes those features.
• The system shall display online test progress, including test name, time taken, score, and status such as not taken, in progress, failed, or passed.
• The system shall store driver comments and lesson start and end times in driver notes.
• The system shall connect with the DMV so DriverPass can receive updates to rules, policies, and sample questions and be notified when updates occur.
• The system shall track who created, modified, or canceled reservations.
• The system shall generate activity reports that can be printed or downloaded.
• The system shall allow authorized users to download reports and selected information for use in tools such as Excel.
• The system shall provide role-based access for customers, the secretary, the owner/management, and the IT officer.

User Interface

• Customer interface: customers should be able to access their account, manage appointments, access applicable online classes and practice tests, view test progress, and review relevant lesson information.
• Secretary interface: the secretary should be able to enter customer information and schedule, cancel, or modify appointments for customers who call or visit the office.
• Owner/management interface: management should be able to view business information, access reports, and review activity records.
• IT officer interface: the IT officer should be able to maintain accounts, reset passwords, block access, and perform system administration tasks.
• The interface should be web-based and accessible from computers and mobile devices.
• The system should include an input form for student/customer information, a contact page, test-progress information, and driver-note information.

Assumptions

• Users will have internet access when they need to create or modify system records.
• DriverPass will provide accurate customer, driver, vehicle, and package information for initial system configuration.
• The DMV will provide an accessible mechanism for transmitting relevant rules, policies, or sample questions.
• Cloud hosting and the supporting database will be available and maintained by the selected service provider.
• The final details for password policy, account lockout thresholds, and exact notification mechanisms will be finalized during detailed design because they are not specified in the interview transcript.

Limitations

• The initial release will not allow a nondeveloper to create or remove entire system modules or training packages; the interview identifies that capability as a future enhancement.

• The transcript does not specify exact performance targets, database technology, cloud vendor, password complexity rules, or brute-force lockout thresholds.
• The system depends on internet connectivity for modifying or updating centralized records.
• The DMV integration depends on the availability and technical capabilities of the external DMV connection.
• The project schedule is constrained by the dates and resource availability discussed in the interview.

Gantt Chart

